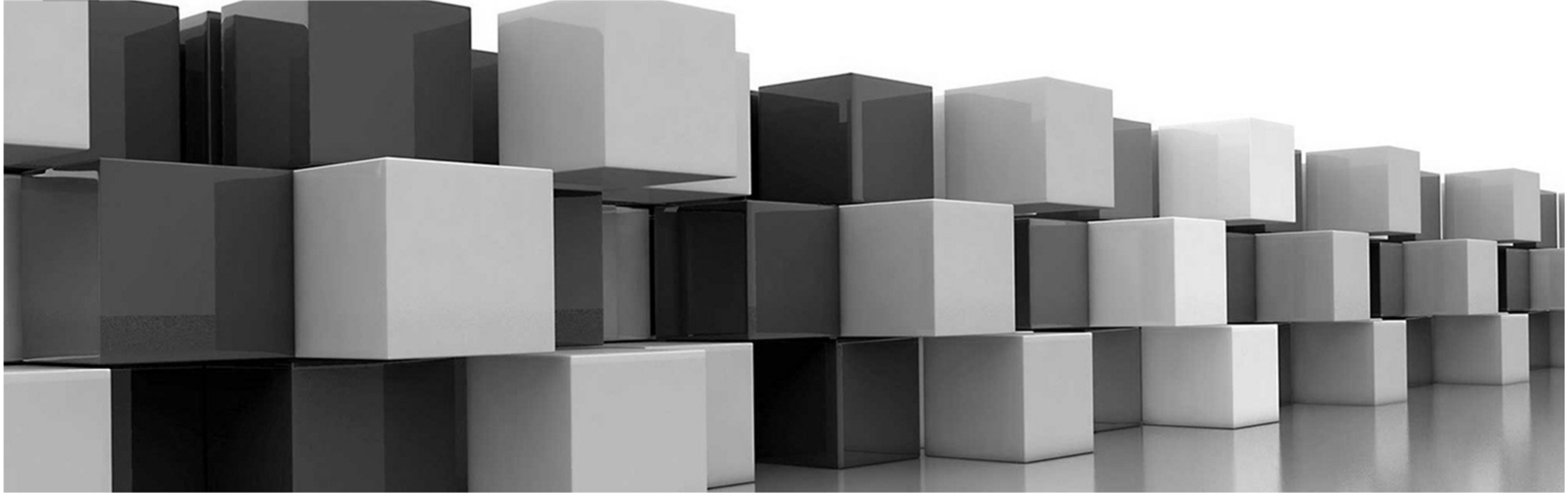


Chapter 7 Symmetric Matrices and Quadratic Forms

林昭宏
GeoAI Lab.
國立成功大學 測量及空間資訊學系

7. Symmetric Matrices and Quadratic Forms

7.1 Diagonalization of symmetric matrices

7.2 Quadratic forms

7.4 The singular value decomposition

Keywords

- Symmetric matrix: 對稱矩陣
- Orthogonal diagonalization: 正交對角化
- Orthonormal diagonalization: 單位正交對角化
- Spectral decomposition: 頻譜分解
- Quadratic form: 二次型
- Covariance matrix: 共變異數矩陣/協方差矩陣
- Positive definite/semidefinite: 正定/半正定
- Singular value decomposition (SVD):奇異值分解

7.1 Diagonalization of symmetric matrices

- Definition
  A symmetric matrix is a matrix **A** such that $\mathbf{A}^T = \mathbf{A}$.

- A symmetric matrix is necessarily square. Its main diagonal entries are arbitrary, but its other entries occur in pairs on opposite sides of the main diagonal.

- Ex.

Symmetric: $\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}, \begin{bmatrix} -1 & 6 & -2 \\ 6 & 2 & 0 \\ -2 & 0 & -3 \end{bmatrix}, \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix}.$

Non-symmetric: $\begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}, \begin{bmatrix} -1 & 6 & -2 \\ 6 & 2 & 0 \\ -3 & 0 & -3 \end{bmatrix}, \begin{bmatrix} a & d & e \\ -d & b & f \\ e & f & c \end{bmatrix}.$

- What is about eigenvectors of a symmetric matrix ?

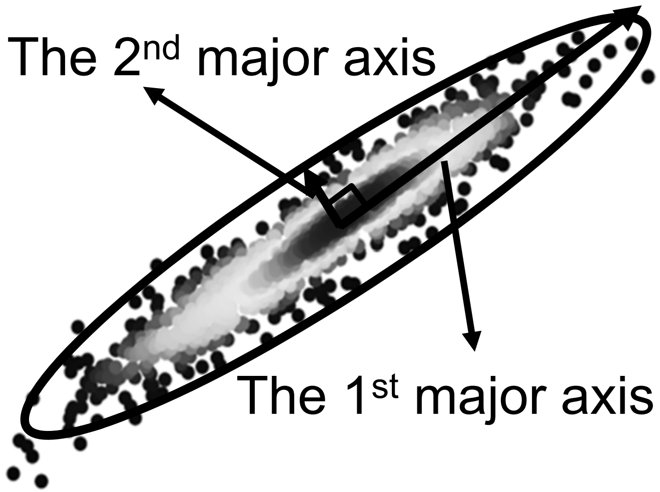
- Theorem 1

  If **A** is symmetric, then any two distinct eigenvectors from different eigenspaces are orthogonal.

  Proof.

  Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be eigenvectors that correspond to distinct eigenvalues λ_1 and λ_2. To show that $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, compute

$$
\begin{aligned}
\lambda_1 \mathbf{v}_1 \cdot \mathbf{v}_2 &= (\lambda_1 \mathbf{v}_1)^T \mathbf{v}_2 = (\mathbf{A}\,\mathbf{v}_1)^T \mathbf{v}_2 \\
&= (\mathbf{v}_1^T \mathbf{A}^T)\,\mathbf{v}_2 = \mathbf{v}_1^T (\mathbf{A}^T \mathbf{v}_2) \\
&= \mathbf{v}_1^T (\mathrm{A}\,\mathbf{v}_2) \\
&= \mathbf{v}_1^T (\lambda_2 \mathbf{v}_2) \\
&= \lambda_2 \mathbf{v}_1^T \mathbf{v}_2 = \lambda_2 \mathbf{v}_1 \cdot \mathbf{v}_2 .
\end{aligned}
$$



  Hence $(\lambda_1 - \lambda_2)\,\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, but $\lambda_1 - \lambda_2 \neq 0$, so $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$.

- A matrix $\mathbf{A}$ is said to be orthogonally diagonalizable if there are an orthonormal matrix $\mathbf{P}$ (with $\mathbf{P}^{-1} = \mathbf{P}^T$) and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{P}\,\mathbf{D}\,\mathbf{P}^T = \mathbf{P}\,\mathbf{D}\,\mathbf{P}^{-1}.$$

- To orthogonally diagonalize a $n \times n$ matrix, we must find n linearly independent and orthonormal eigenvectors. Moreover, if $\mathbf{A}$ is orthogonally diagonalizable as the above equation, then $\mathbf{A}$ is a symmetric matrix

$$\mathbf{A}^T = (\mathbf{P}\,\mathbf{D}\,\mathbf{P}^T)^T = \mathbf{P}^{TT}\mathbf{D}^T\mathbf{P}^T = \mathbf{P}\,\mathbf{D}\,\mathbf{P}^T = \mathbf{A}.$$

- Theorem 2.

An $n \times n$ matrix $\mathbf{A}$ is orthogonally diagonalizable if and only if $\mathbf{A}$ is a symmetric matrix.

Proof.

The proof is much harder and is omitted.

Easy orthogonally diagonalizable $\Rightarrow$ symmetric

Hard orthogonally diagonalizable $\Leftarrow$ symmetric

Ex.

Orthogonally diagonalize the matrix $\mathbf{A} = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$.

Solution.

$$\lambda_1 = 7\,,7:\ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},\ \mathbf{v}_2 = \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix};\ \ \lambda_2 = -2:\ \mathbf{v_3} = \begin{bmatrix} -1 \\ -1/2 \\ 1 \end{bmatrix}.$$

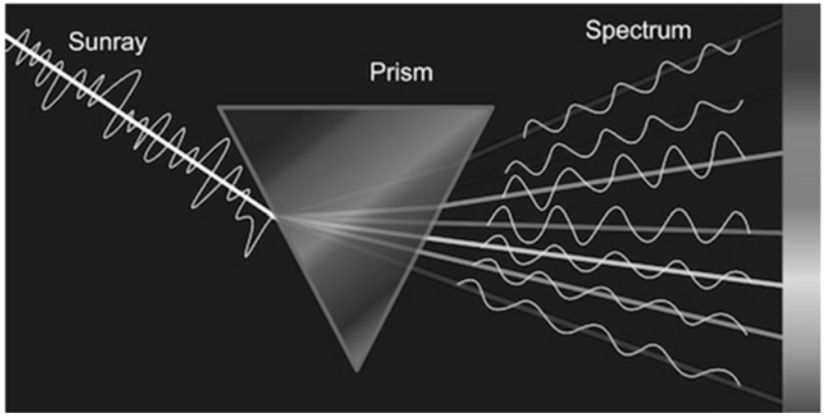
Although $\mathbf{v}_1$ and $\mathbf{v}_2$ are only linearly independent, not orthogonal; but they can be become orthogonal by the Gram-Schmidt process. And, normalize $\mathbf{v}_3$.

$$\mathbf{v}_1 \Rightarrow \mathbf{u}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix},\ \mathbf{v}_2 \Rightarrow \mathbf{u_2} = \begin{bmatrix} -1/\sqrt{18} \\ 4/\sqrt{18} \\ 1/\sqrt{18} \end{bmatrix},\ \mathbf{v}_3 \Rightarrow \mathbf{u_3} = \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}.$$

Then $\mathbf{P} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{18} & -2/3 \\ 0 & 4/\sqrt{18} & -1/3 \\ 1/\sqrt{2} & 1/\sqrt{18} & 2/3 \end{bmatrix},\quad \mathbf{D} = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$

The spectral theorem

- The set of eigenvalues of a matrix **A** is sometimes called the *spectrum* of **A**, and the following description of the eigenvalues is called a *spectral theorem*.



- Theorem 3. (*The spectral theorem for symmetric matrices*)
  A $n \times n$ symmetric matrix **A** has the following properties:

  a. **A** has *n* real eigenvalues, counting multiplicities.

  b. The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.

  c. The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal.

  d. **A** is orthogonally diagonalizable.

Spectral decomposition

- Support $\mathbf{A} = \mathbf{PDP}^{-1}$, where the columns of $\mathbf{P}$ are orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of $\mathbf{A}$ and the corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$ are in the diagonal matrix $\mathbf{D}$. Then, since $\mathbf{P}^{-1} = \mathbf{P}^T$,

$$\mathbf{A} = \mathbf{PDP}^T = [\mathbf{u}_1 \cdots \mathbf{u}_n]\begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_n \end{bmatrix}\begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} = [\lambda_1\mathbf{u}_1 \cdots \lambda_n\mathbf{u}_n]\begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix}.$$

Using the column-row expansion of a product, we can write

$$\mathbf{A} = \lambda_1\mathbf{u}_1\mathbf{u}_1^T + \lambda_2\mathbf{u}_2\mathbf{u}_2^T + \cdots + \lambda_n\mathbf{u}_n\mathbf{u}_n^T$$

$$\boxed{\mathbf{A}} = \lambda_1 \times \boxed{\mathbf{u}_1\mathbf{u}_1^T} + \lambda_2 \times \boxed{\mathbf{u}_2\mathbf{u}_2^T} + \cdots + \lambda_n \times \boxed{\mathbf{u}_n\mathbf{u}_n^T}$$

This representation of $\mathbf{A}$ is called a spectral decomposition of $\mathbf{A}$. Each term in the above equation is a $n \times n$ matrix. Each matrix $\mathbf{u}_j\mathbf{u}_j^T$ is a projection matrix such that $(\mathbf{u}_j\mathbf{u}_j^T)\mathbf{x}$ is the orthogonal projection of $\mathbf{x}$ onto the subspace spanned by $\mathbf{u}_j$.

❁ Ex.
Construct a spectral decomposition of the matrix **A** that has the orthogonal diagonalization

$$\mathbf{A} = \begin{bmatrix} 7 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}.$$

Solution.
Denote the columns of **P** by $\mathbf{u}_1$ and $\mathbf{u}_2$. Then

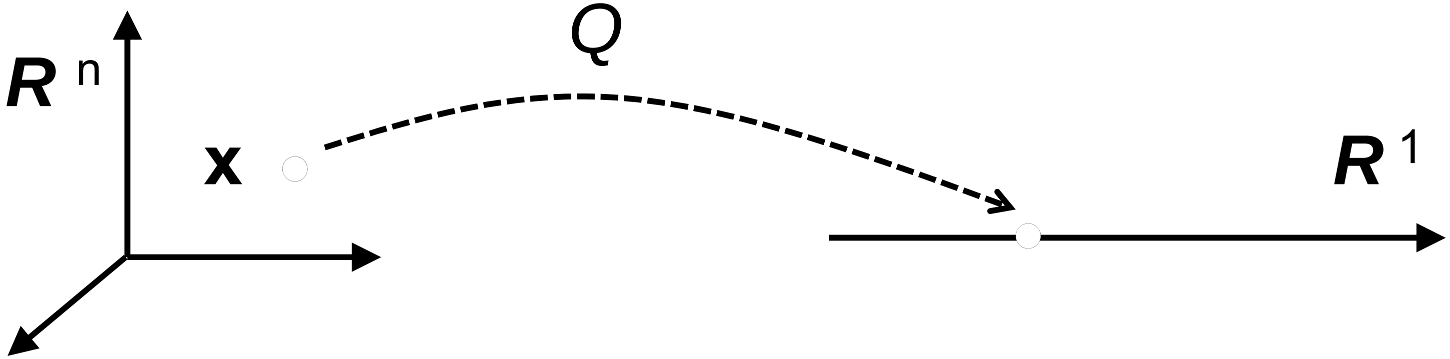
$$\mathbf{A} = 8\mathbf{u}_1\mathbf{u}_1^T + 3\mathbf{u}_2\mathbf{u}_2^T$$

$$= 8\begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} + 3\begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$= 8\begin{bmatrix} 4/5 & 2/5 \\ 2/5 & 1/5 \end{bmatrix} + 3\begin{bmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{bmatrix}.$$

7.2 Quadratic forms

- A quadratic form on $\boldsymbol{R}^n$ is a function Q defined on $\boldsymbol{R}^n$ whose value at a vector $\mathbf{x}$ in $\boldsymbol{R}^n$ can be computed by an expression of the form where $\mathbf{A}$ is an nxn symmetric matrix. The matrix $\mathbf{A}$ is called the matrix of the quadratic form.



- The simplest example of a nonzero quadratic form is

$$Q(\mathbf{x}) = \mathbf{x}^T \mathbf{I}\, x = \|\mathbf{x}\|^2.$$

- Ex. For **x** in $\boldsymbol{R}^2$,

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 4x_1^2 + 3x_2^2.$$

$$\begin{aligned} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 3x_1 - 2x_2 & -2x_1 + 7x_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= 3x_1^2 - 2x_1x_2 - 2x_1x_2 + 7x_2^2 \\ &= 3x_1^2 - 4x_1x_2 + 7x_2^2. \end{aligned}$$

Ex. For **x** in $\boldsymbol{R}^3$,

$$\begin{aligned} &5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3 \\ &= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 5 & -1/2 & 0 \\ -1/2 & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}. \end{aligned}$$

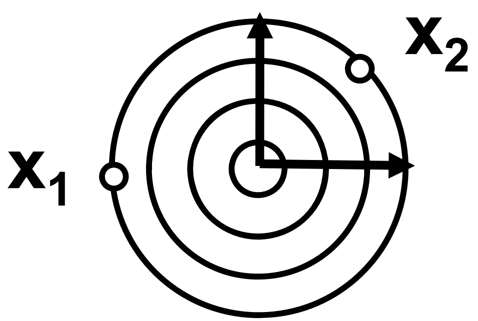
Geometric meaning of quadratic forms $Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A}\, \mathbf{x}$

$\mathbf{x}^T\mathbf{A}\mathbf{x}$ means the **distance** of $\mathbf{x}$ in the quadratic form.

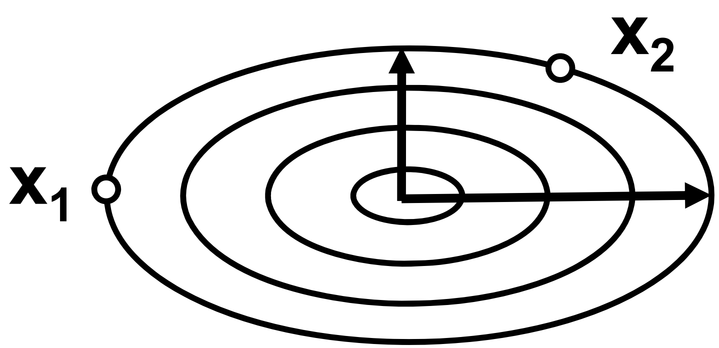
q的概念是距離

❁ Ex.

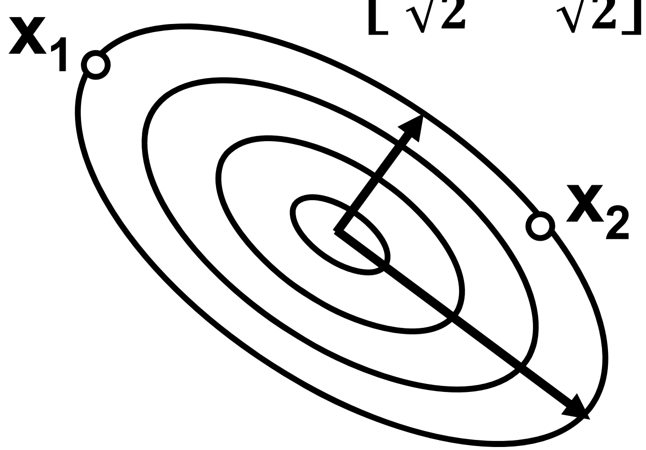
$$Q_1(\mathbf{x}) = \mathbf{x}^T\, \mathbf{I}\, \mathbf{x} \qquad Q_2(\mathbf{x}) = \mathbf{x}^T \begin{bmatrix} 1.5 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x} \qquad Q_3(\mathbf{x}) = \mathbf{x}^T \begin{bmatrix} \frac{3}{2\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \mathbf{x}$$



Distance field of $Q_1(\mathbf{x})$



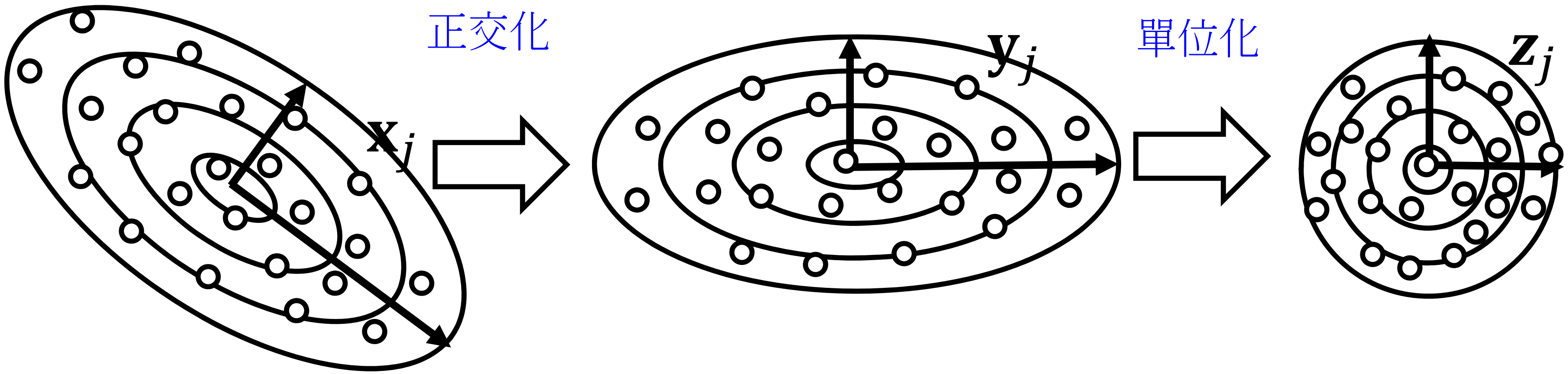
Distance field of $Q_2(\mathbf{x})$



Distance field of $Q_3(\mathbf{x})$

Applications of $Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A}\, \mathbf{x}$

Given a point set $(\mathbf{x}_1, \cdots, \mathbf{x}_n)$ and calculate the covariance matrix Σ of this point set.



$Q(\mathbf{x}) = \mathbf{x}^T \Sigma\, \mathbf{x}$ $\qquad$ $Q(\mathbf{y}) = \mathrm{y}^T \mathbf{D}\, \mathbf{y}$ $\qquad$ $Q(\mathrm{z}) = \mathrm{z}^T \mathbf{I}\, \mathbf{z}$

The location of $\mathbf{x}_j = (2, 0)$ in the point set?

Application of quadratic forms $Q(\mathbf{x}) = \mathbf{x}^T \Sigma \mathbf{x}$

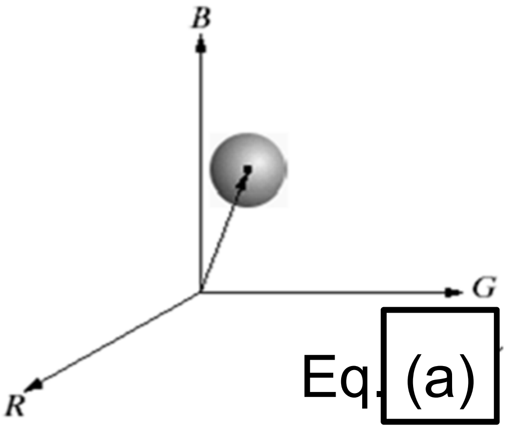
二次式 應用一：依照顏色劃分圖片

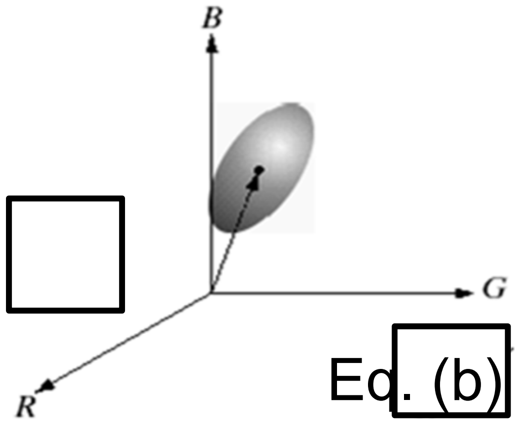
- Image segmentation in RGB vector space: **Partition an image into regions according to its colors**.
- The measurement of **color similarity** is the Euclidean distance between two colors $\boldsymbol{z}$ and $\boldsymbol{a}$.

$$D(\mathbf{z},\mathbf{a}) = \|\mathbf{z}-\mathbf{a}\| \tag{a}$$

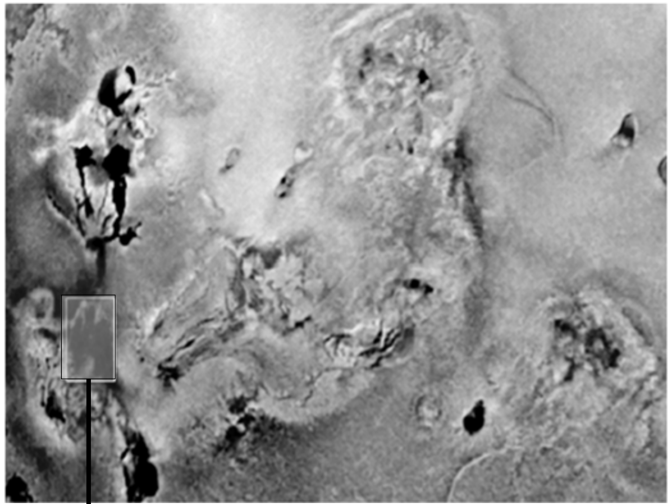
- A useful generalization of Eq. (a) is a distance measure of the form

$$D(\mathbf{z},\mathbf{a}) = [(\mathbf{z}-\mathbf{a})^T \Sigma (\mathbf{z}-\mathbf{a})] \tag{b}$$





Application of quadratic forms $Q(\mathbf{x}) = \mathbf{x}^T \Sigma \mathbf{x}$



Calculate Σ of the pixels in 

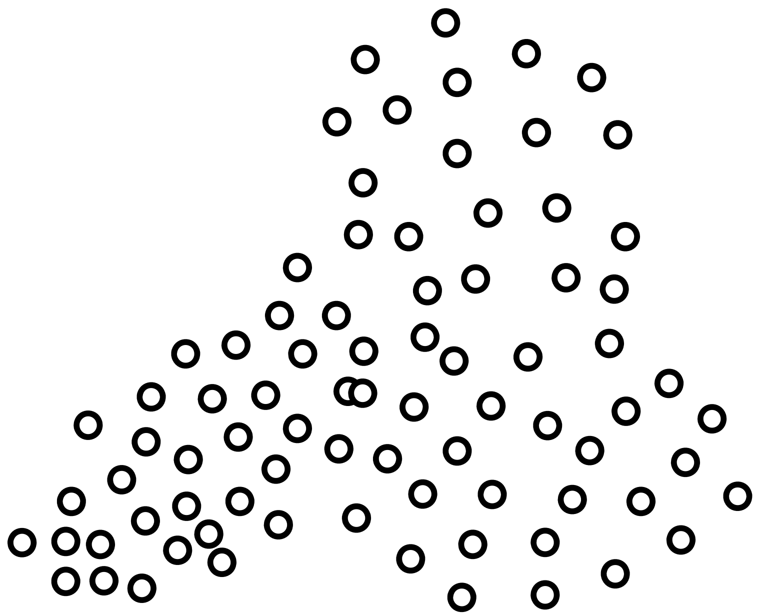
Selected by users

$Q(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\Sigma}\, \mathbf{x} <$ threshold

Applications of $Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$

二次式 應用二：資料分群

Given a point set $(\mathbf{x}_1, \cdots, \mathbf{x}_n)$ and classify the dataset into three groups using Gaussian mixture model.



Applications of $Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A}\, \mathbf{x}$

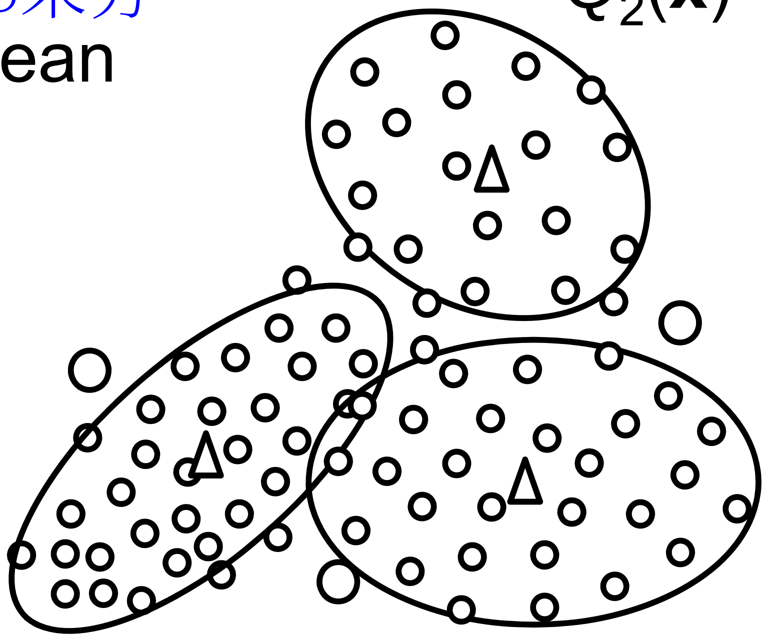
Given a point set $(\mathbf{x}_1, \cdots, \mathbf{x}_n)$ and classify the dataset into three groups using Gaussian mixture model.

資料分群

用 sigma1,2,3來分

Δ : mean

$Q_2(\mathbf{x}) = \mathbf{x}^T \Sigma_2\, \mathbf{x}$



Samples "○" belong to which group?

Check the distances

$Q_1(○)$, $Q_2(○)$, $Q_3(○)$

離群值x: 計算q1(x),q2(x),q3(x)

分到距離最近的群

$Q_1(\mathbf{x}) = \mathbf{x}^T \Sigma_1\, \mathbf{x}$

$Q_3(\mathbf{x}) = \mathbf{x}^T \Sigma_3\, \mathbf{x}$

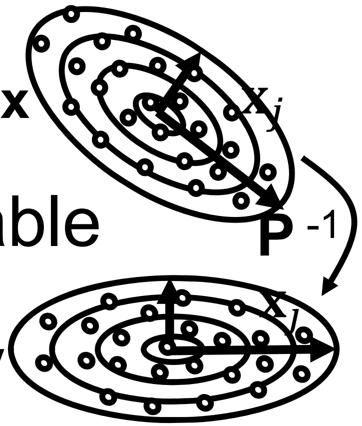
變數變換

Change of variable in a quadratic form

- If $\mathbf{x}$ is a variable vector in $\boldsymbol{R}^n$, then a change of variable is an equation of the form

$$\mathbf{x} = \mathbf{P}\,\mathbf{y} \quad \text{or} \quad \mathbf{y} = \mathbf{P}^{-1}\mathbf{x},$$

where $\mathbf{P}$ is an invertible matrix and $\mathbf{y}$ is a new variable vector in $\boldsymbol{R}^n$.



- The $\mathbf{y}$ is the coordinate vector of $\mathbf{x}$ relative to the basis of $\boldsymbol{R}^n$ determined by the columns of $\mathbf{P}$ ($[\mathbf{x}]_E = \mathbf{P}_\beta\,[\mathbf{x}]_\beta$).

- If the change of variable is substituted into a quadratic form $\mathbf{x}^T\mathbf{A}\,\mathbf{x}$, then

$$\mathbf{x}^T\mathbf{A}\,\mathbf{x} = (\mathbf{P}\,\mathbf{y})^T\mathbf{A}\,(\mathbf{P}\,\mathbf{y}) = \mathbf{y}^T\mathbf{P}^T\mathbf{A}\,\mathbf{P}\,\mathbf{y} = \mathbf{y}^T(\mathbf{P}^T\mathbf{A}\,\mathbf{P})\,\mathbf{y}.$$

Since $\mathbf{A}$ is orthogonally diagonalizable (Theorem 2), there is a orthonormal matrix $\mathbf{P}$ such that $\mathbf{P}^T\mathbf{A}\,\mathbf{P}$ is a diagonal matrix $\mathbf{D}$ and thus, the quadratic form becomes

- Ex.

  Make a change of variable that transforms a quadratic form with **A**=$\begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$ into a quadratic form with no cross-product term.

  Solution.

  Step 1. Find eigenvalues and eigenvectors of **A**

$$\lambda_1 = 3:\ \mathbf{v}_1 = \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix};\quad \lambda_2 = -7:\ \mathbf{v_2} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix};$$

  Step 2. Construct **P** and **D**,

$$\mathbf{P} = \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & -7 \end{bmatrix}$$

  Step 3. Transform quadratic form with no cross-product term

$$\mathbf{x}^T \mathbf{A}\,\mathbf{x} = \mathbf{y}^T (\mathbf{P}^T \mathbf{A}\,\mathbf{P})\,\mathbf{y} = \mathbf{y}^T \mathbf{D}\mathbf{y} = 3y_1^2 - 7y_2^2.$$

Ex.

Make a change of variable that transforms a quadratic form with $Q(\mathbf{x}) = x_1^2 - x_3^2 - 4x_1x_2 + 4x_2x_3$ into a quadratic form with no cross-product term.

Solution.

$$Q = \mathbf{x}^T\mathbf{A}\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Step 1. Find eigenvalues and eigenvectors of **A**

$$\begin{vmatrix} 1-\lambda & -2 & 0 \\ -2 & 0-\lambda & 2 \\ 0 & 2 & -1-\lambda \end{vmatrix} = \lambda^3 - 9\lambda = \lambda(\lambda+3)(\lambda-3) = 0$$

$$\lambda_1 = 0:\ \mathbf{v}_1 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix};\ \lambda_2 = -3:\ \mathbf{v_2} = \begin{bmatrix} -1/3 \\ -2/3 \\ 2/3 \end{bmatrix};\ \lambda_3 = 3:\ \mathbf{v_3} = \begin{bmatrix} -2/3 \\ 2/3 \\ 1/3 \end{bmatrix}.$$

- Ex.

Step 2. Construct **P** and **D**,

$$\mathbf{P} = \begin{bmatrix} 2/3 & -1/3 & -2/3 \\ 1/3 & -2/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Step 3. Transform quadratic form with no cross-product term

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{y}^T (\mathbf{P}^T \mathbf{A} \mathbf{P}) \mathbf{y} = \mathbf{y}^T \mathbf{D} \mathbf{y} = -3y_2^2 + 2y_3^2.$$

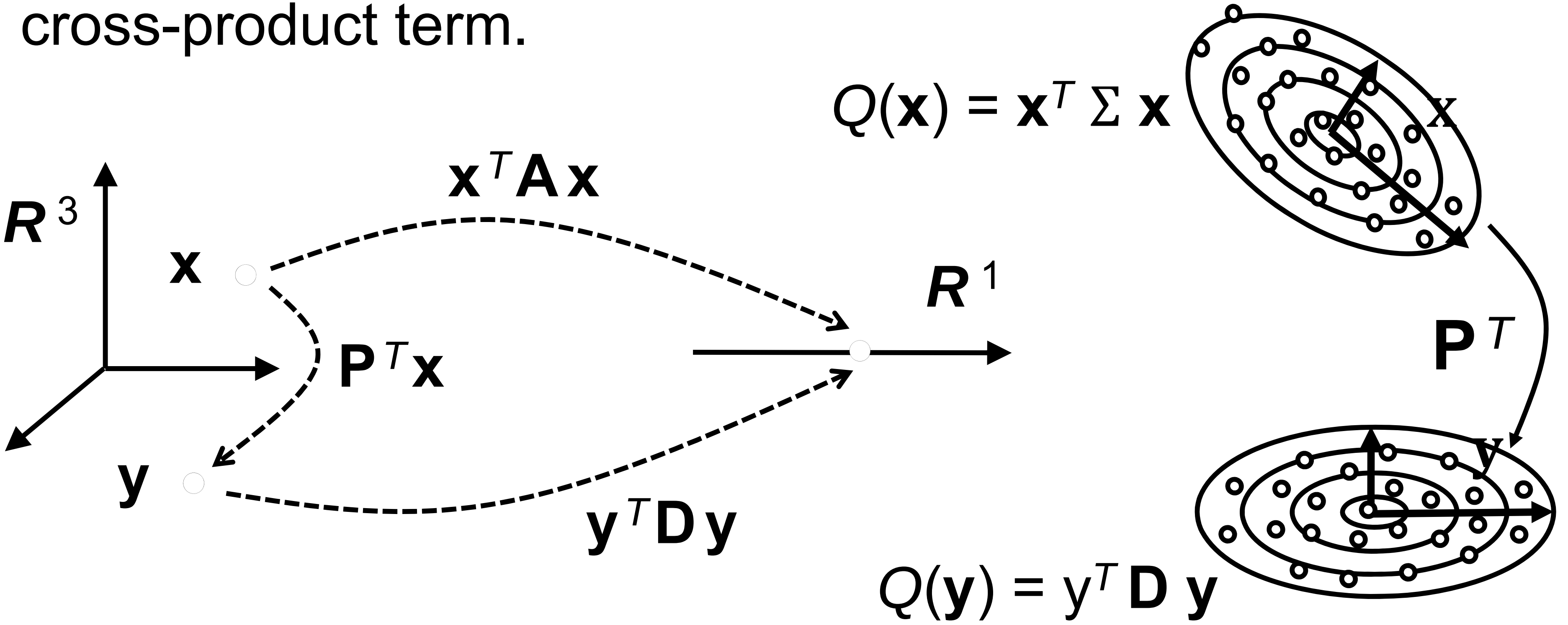
In-class Exercise #15

Q. Find a change of variable that removes the cross-product term from $Q(\mathbf{x}) = 5x_1^2 - 4x_1x_2 + 5x_2^2$.

Answer.

❁ Theorem 4. (The principal axes theorem) 主軸原理

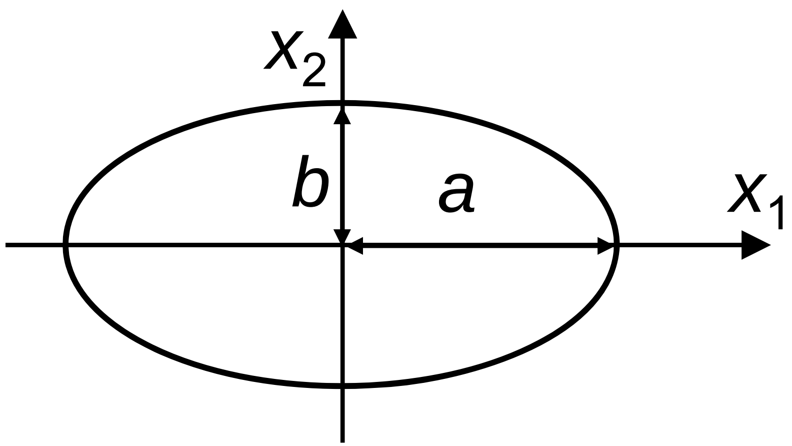
Let $\mathbf{A}$ be an $n\times n$ symmetric matrix. Then there is an orthogonal change of variable, $\mathbf{x} = \mathbf{P}\,\mathbf{y}$, that transforms the quadratic form $\mathbf{x}^T\mathbf{A}\,\mathbf{x}$ into a quadratic form $\mathbf{y}^T\mathbf{D}\,\mathbf{y}$ with no cross-product term.

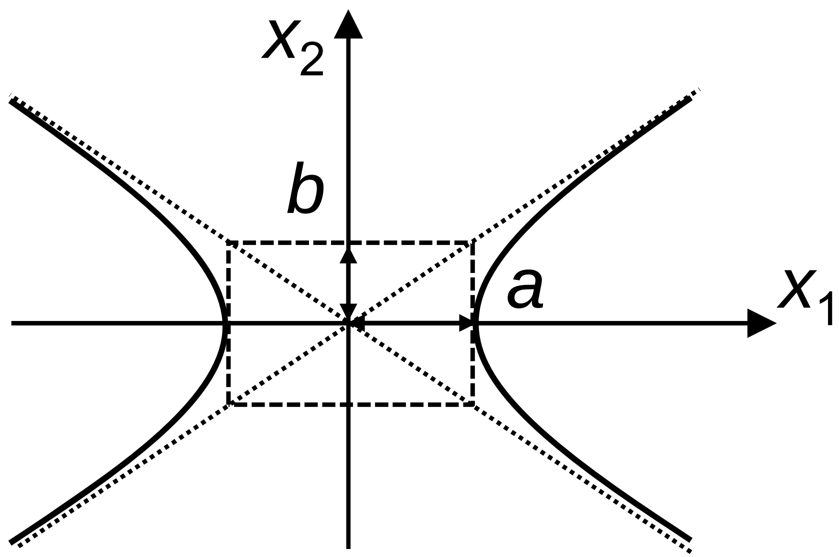


❁ The columns of $\mathbf{P}$ in Theorem 4 are called $\boxed{}$ of the quadratic form $\mathbf{x}^T\mathbf{A}\,\mathbf{x}$.

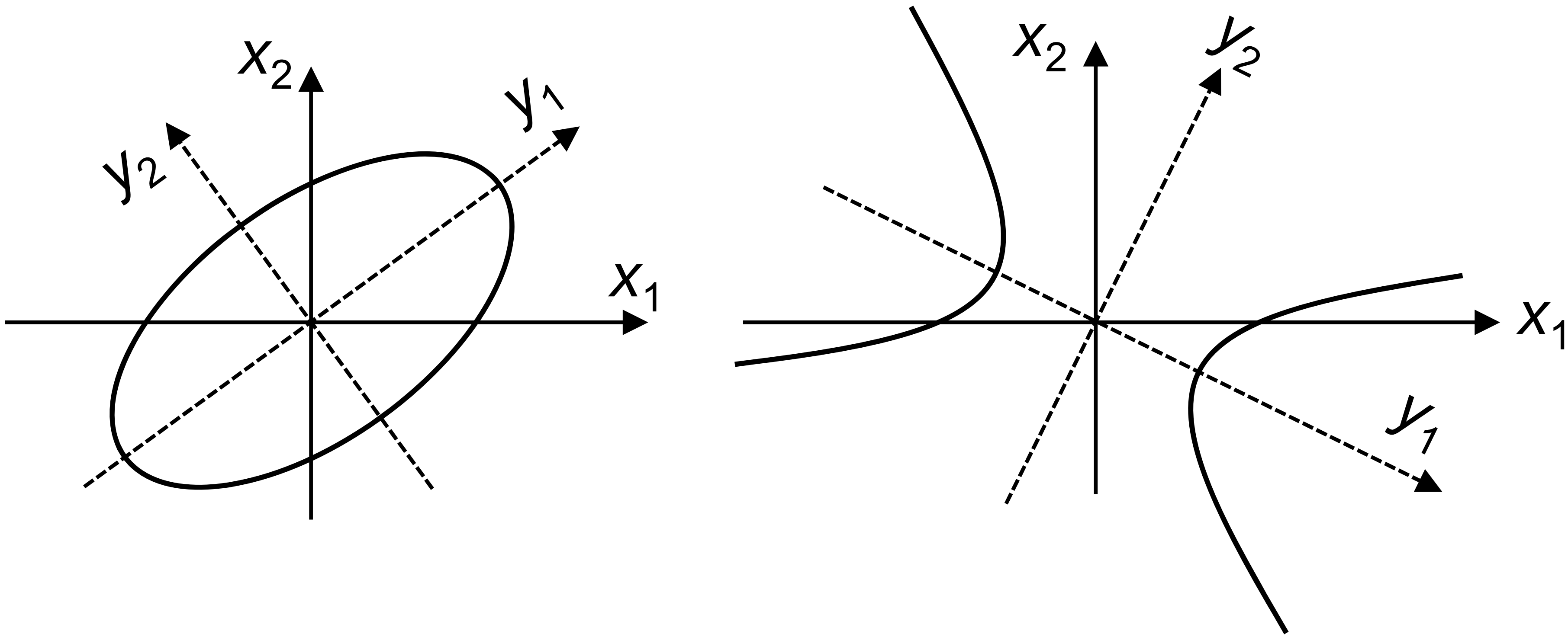
A geometric view of principal axes

- Support $Q(\mathbf{x}) = \mathbf{x}^T\mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is an invertible 2×2 symmetric matrix and let c be a constant. Then any $\mathbf{x}$ in $\boldsymbol{R}^2$ that satisfy $\mathbf{x}^T\mathbf{A}\mathbf{x} = c$ corresponds to
  i. a circle,
  ii. an ellipse,
  iii. a hyperbola,





- If $\mathbf{A}$ is a diagonal matrix, the graph of $\mathbf{x}^T\mathbf{A}\mathbf{x} = c$ is in standard position.
  If $\mathbf{A}$ is not a diagonal matrix, the graph of $\mathbf{x}^T\mathbf{A}\mathbf{x} = c$ is rotated out of standard position.



An ellipse and a hyperbola are not in standard position (out of standard position).

Classifying quadratic forms

- When $\mathbf{A}$ is an $n \times n$ matrix, the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ is a real-values function with $\boldsymbol{R}^n$.

- Definition. A quadratic form Q is:
  a. positive definite (正定) if $Q(\mathbf{x}) > 0$ for all $\mathbf{x} \neq 0$,
  b. negative definite (負定) if $Q(\mathbf{x}) < 0$ for all $\mathbf{x} \neq 0$,
  c. indefinite if $Q(\mathbf{x})$ has both positive and negative values,
  d. positive semidefinite (半正定) if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \neq 0$,
  e. negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x} \neq 0$.

- **Theorem 5. (Quadratic forms and eigenvalues)**
  Let **A** be an $n \times n$ symmetric matrix. Then a quadratic form $\mathbf{x}^{\mathrm{T}}\mathbf{A}\mathbf{x}$ is:
  a. positive definite if and only if the eigenvalues of **A** are all positive.
  b. negative definite if and only if the eigenvalues of **A** are all negative.
  c. indefinite if and only if **A** has both positive and negative eigenvalues.

- Ex. Is $Q(\mathbf{x}) = 5x_1^2 + 2x_2^2 - x_3^2 + 4x_1x_2 + 4x_2x_3$ positive definite?
  Solution.
  $$Q(\mathbf{x}) = \mathbf{x}^T\mathbf{A}\mathbf{x} = [x_1 \quad x_2 \quad x_3]\begin{bmatrix} 5 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$
  The eigenvalues of **A** are 5, 2, and -1. So *Q* is an indefinite quadratic form.

7.4 The singular value decomposition

- Eigen-analysis can be applied to a squared matrix $\mathbf{A}_{n\times n}$, that is,

$$\mathbf{Av} = \lambda\mathbf{v}$$

- A non-squared matrix $\mathbf{A}_{m\times n}$ also has similar eigen formula, that is,

singular value

$$\mathbf{Av} = \sigma\mathbf{u}$$

non-squared matrix singular vector

- In this subsection,
  - for $n > m$ matrices ( ) : ||**Ax**|| optimization
  - for all non-squared matrices : SVD
  - for $m > n$ matrices ( ) : Pseudo-inverse PCA

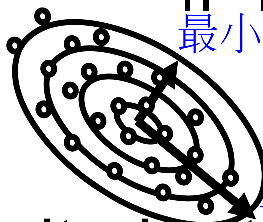
7.4 The singular value decomposition

- Not all matrices can be factored as $\mathbf{A}=\mathbf{PDP}^{-1}$ with $\mathbf{D}$ diagonal; however, a factorization $\mathbf{A}= \mathbf{U}\Sigma\mathbf{V}^T$ is possible for any $m\times n$ matrix $\mathbf{A}$. A special factorization of this type, called the singular value decomposition, is one of the most useful matrix factorizations in applied linear algebra.

- The principal of the singular value decomposition

The absolute values of the eigenvalues of a symmetric matrix $\mathbf{B}$ measure the amounts that $\mathbf{B}$ stretches or shrinks certain vectors (the eigenvectors). If $\mathbf{Bx} = \lambda\mathbf{x}$ and $||\mathbf{x}|| = 1$, then 伸縮長度由特徵值決定

$$||\mathbf{Bx}|| = ||\lambda\mathbf{x}|| = |\lambda|\,||\mathbf{x}|| = |\lambda|.$$

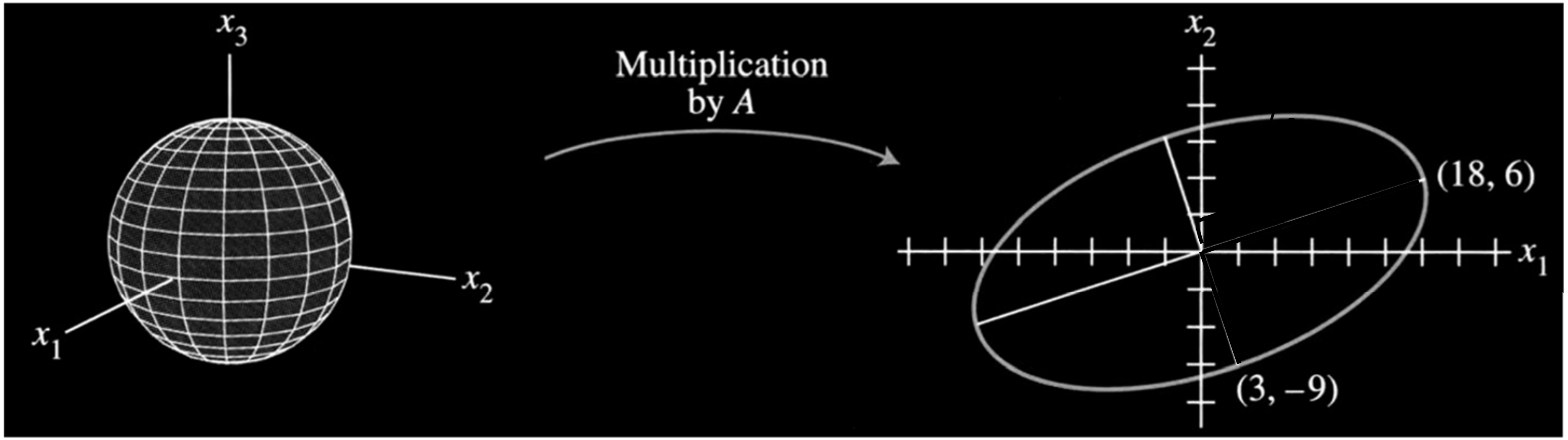


If λ_1 is the eigenvalue with the greatest magnitude, then a corresponding unit eigenvector $\mathbf{v}_1$ identifies a direction in which the stretching effect of $\mathbf{B}$ is greatest.

||Ax|| optimization

❁ Ex.

If $\mathbf{A} = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$, then the linear transformation $\mathbf{x} \mapsto \mathbf{Ax}$ maps the unit sphere $\{\mathbf{x} : \|\mathbf{x}\| = 1\}$ in $\boldsymbol{R}^3$ onto an ellipse in $\boldsymbol{R}$.



Find a unit vector $\mathbf{x}$ at which the length $\|\mathbf{Ax}\|$ is maximized, and compute this maximum length.

Solution.

The quantity $\|\mathbf{Ax}\|^2$ is maximized at the same $\mathbf{x}$ that maximizes $\|\mathbf{Ax}\|$, and $\|\mathbf{Ax}\|^2$ is easier to study. symmetric matrix

eigenvalue of $\mathbf{A}^T\mathbf{A}$

$$\|\mathbf{Ax}\|^2 = (\mathbf{Ax})^T(\mathbf{Ax}) = \mathbf{x}^T\mathbf{A}^T\mathbf{Ax} = \mathbf{x}^T(\underline{\mathbf{A}^T\mathbf{A}})\mathbf{x} = \mathbf{x}^T\,\underline{\lambda}\,\mathbf{x} = \lambda$$

$\mathbf{A}^T\mathbf{A}$ is a symmetric matrix, since $(\mathbf{A}^T\mathbf{A})^T = \mathbf{A}^T\mathbf{A}^{TT} = \mathbf{A}^T\mathbf{A}$. So the problem now is to maximize the quadratic form $\mathbf{x}^T(\mathbf{A}^T\mathbf{A})\mathbf{x}$ subject to the constrain $||\mathbf{x}|| = 1$.

The maximum value is the greatest eigenvalue λ_1 of $\mathbf{A}^T\mathbf{A}$. Also, the maximum value is attained at a unit eigenvector of $\mathbf{A}^T\mathbf{A}$ corresponding to λ_1.

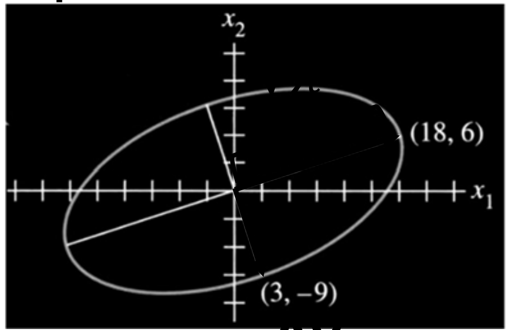
$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} = \begin{bmatrix} 80 & 100 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{bmatrix}.$$

The eigenvalues of $\mathbf{A}^T\mathbf{A}$ are $\lambda_1 = 360$, $\lambda_2 = 90$, and $\lambda_3 = 0$.

The corresponding unit eigenvectors are:

$$\mathbf{v}_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}, \text{ and } \mathbf{v}_3 = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}.$$

The maximum value of $||\mathbf{Ax}||^2$ is 360, attained when $\mathbf{x}$ is the unit vector $\mathbf{v}_1$. The vector $\mathbf{Av}_1$ is a point on the ellipse in the above figure farthest from the origin, namely,



$$\mathbf{Av}_1 = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 18 \\ 6 \end{bmatrix}.$$

For $\|\mathbf{x}\| = 1$, the maximum value of $||\mathbf{Ax}||$ is $||\,\mathbf{Av}_1||= \sqrt{360} = 6\sqrt{10}$.

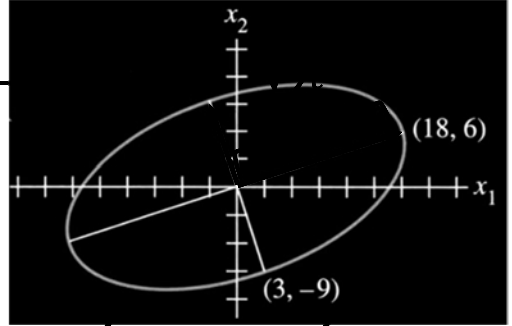
For the second unit eigenvector, $\mathbf{v}_2$,

$$\mathbf{Av}_2 = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \end{bmatrix}.$$

Since $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal; $\mathbf{A}^T\mathbf{A}\mathbf{v_1} = \lambda_1\mathbf{v}_1$ and $\mathbf{A}^T\mathbf{A}\mathbf{v}_2 = \lambda_2\mathbf{v_2}$.

$$\mathbf{Av}_1 \cdot \mathbf{A}\mathbf{v_2} = (\mathbf{Av}_1)^{\boldsymbol{T}}\mathbf{Av}_2 = \mathbf{v}_1^T\mathbf{A}^T\mathbf{Av}_2 = \mathbf{v}_{\mathbf{1}}^T(\mathbf{A}^T\mathbf{A}\mathbf{v_2}) = \mathbf{v}_1^T\lambda_2\mathbf{v}_2$$
$$= \lambda_2\mathbf{v}_1^T\mathbf{v}_2 = \mathbf{0}.$$

$\Rightarrow$ $\mathbf{Av}_1$ and $\mathbf{Av}_2$ are orthogonal.



The singular values of an $m\times n$ matrix

求最佳解：1. regression least square solution, 2. PCA, 3.SVD(二次式)算最小距離

- Let **A** be an $m\times n$ matrix. Then $\mathbf{A}^T\mathbf{A}$ is a symmetric and can be orthogonally diagonalized. Let $\{\mathbf{v}_1,\ldots,\mathbf{v}_n\}$ be an orthonormal basis for $\boldsymbol{R}^n$ consisting of eigenvectors of $\mathbf{A}^T\mathbf{A}$, and let $\lambda_1, \ldots, \lambda_n$ be the associated eigenvalues of $\mathbf{A}^T\mathbf{A}$. Then for $1 \le i \le n$,

$$||\mathbf{Av}_i||^2 = (\mathbf{Av}_i)^T\mathbf{Av}_i = \mathbf{v}_i^T\,\mathbf{A}^T\mathbf{Av}_i = \mathbf{v}_i^T(\lambda_i\,\mathbf{v}_i) = \lambda_i. \qquad (2)$$

So the eigenvalues of (A)TA are all nonnegative

By renumbering, we can get $\lambda_1 \ge \lambda_2 \ge \ldots \ge \lambda_n \ge 0$ (mean that all $\mathbf{A}^T\mathbf{A}$ **are** positive semidefinite(半正定)). 在這我們討論的都是：半正定

- The *singular values* of **A** are the square roots of the eigenvalues of $\mathbf{A}^T\mathbf{A}$, denoted by $\sigma_1, \ldots, \sigma_n$, and they are arranged in decreasing order. That is, $\sigma_i = \sqrt{\lambda_i}$. The singular values of **A** are the lengths of the vectors $\mathbf{Av}_1, \ldots, \mathbf{Av}_n$.

The singular value decomposition

SVD

- The decomposition of **A** involves an $m \times n$ "diagonal" matrix Σ of the form

$$\mathbf{\Sigma} = \begin{bmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{3}$$

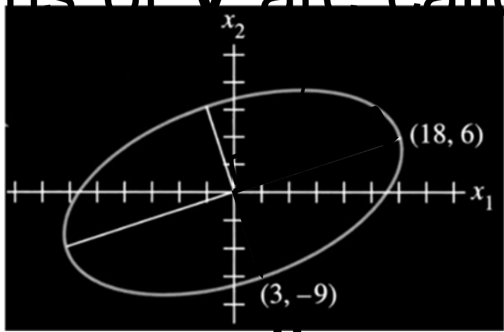
where **D** is an $r \times r$ diagonal matrix for some r not exceeding the smaller of m and n.

- Theorem 10. (The singular value decomposition)
Let **A** be an $m \times n$ matrix with rank r. Then there exists an $m \times n$ matrix Σ as in Eq.(3) for which the diagonal entries in **D** are the first r singular values of **A**, $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0$, and there exist an $m \times m$ orthonormal matrix **U** and an $n \times n$ orthonormal matrix **V** such that

$$\mathbf{A} = \mathbf{U} \Sigma \mathbf{V}^T.$$

- Any factorization **A** = **U** ∑ **V**^{T}, with **U** and **V** orthonormal, ∑ as in Eq.(3), and positive diagonal entries in **D**, is called a singular value decomposition (or *SVD*) of **A**.

- The columns of **U** in such a decomposition are called left singular vectors of **A**, and the columns of **V** are called right singular vectors of **A**.



$\mathbf{u}_2$

Explanations of Theorem 10.

Let λ_i and $\mathbf{v}_i$ be a corresponding eigenvalue and eigenvector,

Normalize each $\mathbf{Av}_i$ to obtain an orthonormal basis $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$,

where $\mathbf{u}_i = \frac{1}{\|\mathbf{Av}_i\|}\mathbf{Av}_i = \frac{1}{\sigma_i}\mathbf{Av}_i$ (4)

$\Rightarrow$ $\mathbf{Av}_i = \sigma_i\mathbf{u}_i$, $1 \leq i \leq r$. ($\mathbf{u}_i \in R^2$, $\mathbf{v}_i \in R^3$ in this example)

$\Rightarrow$ $\mathbf{A} = \mathbf{U} \sum \mathbf{V}^T$, where $\mathbf{U}=[\mathbf{u}_1 \ldots \mathbf{u}_r]$, $\sum=[\mathbf{D}, \mathbf{0}]$, $\mathbf{V}=[\mathbf{v}_1 \ldots \mathbf{v}_r]$

Ex.

Using the results of Ex. in Page 31 to construct a singular value decomposition of $\mathbf{A} = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$.

Solution.

A construction can be divided into three steps.

Step 1. Find an orthogonal diagonalization of $\mathbf{A}^T\mathbf{A}$. That is to find the eigenvalues of $\mathbf{A}^T\mathbf{A}$ and a corresponding orthonormal set of eigenvectors.

Step 2. Set up **V** and $\mathbf{\Sigma}$.

$$\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} 1/3 & -2/3 & 2/3 \\ 2/3 & -1/3 & -2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} 6\sqrt{10} & 0 \\ 0 & 3\sqrt{10} \end{bmatrix}, \mathbf{\Sigma} = [\mathbf{D} \quad 0] = \begin{bmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{bmatrix}$$

Step 3. Construct **U**.

$$\mathbf{u_1} = \frac{1}{\sigma_1}\mathbf{Av_1} = \frac{1}{6\sqrt{10}}\begin{bmatrix}18\\6\end{bmatrix} = \begin{bmatrix}3/\sqrt{10}\\1/\sqrt{10}\end{bmatrix},$$

$$\mathbf{u_2} = \frac{1}{\sigma_2}\mathbf{Av_2} = \frac{1}{3\sqrt{10}}\begin{bmatrix}3\\-9\end{bmatrix} = \begin{bmatrix}1/\sqrt{10}\\-3/\sqrt{10}\end{bmatrix},$$

and $\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{u}_2]$.

Thus the singular value decomposition of **A** is

$$\mathbf{A} = \mathbf{U} \sum \mathbf{V}^T =$$

$$\begin{bmatrix}3/\sqrt{10} & 1/\sqrt{10}\\1/\sqrt{10} & -3/\sqrt{10}\end{bmatrix}\begin{bmatrix}6\sqrt{10} & 0 & 0\\0 & 3\sqrt{10} & 0\end{bmatrix}\begin{bmatrix}1/3 & 2/3 & 2/3\\-2/3 & -1/3 & 2/3\\2/3 & -2/3 & 1/3\end{bmatrix}$$

- Ex.
  Find a singular value decomposition of $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$.

  Solution.

Step 1. Find the eigenvalues of $\mathbf{A}^T\mathbf{A}$ and a corresponding orthonormal set of eigenvectors.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}.$$

$$\lambda_1 = 18\text{: } \mathbf{v}_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}; \ \lambda_2 = 0\text{: } \mathbf{v_2} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

Step 2. Set up **V** and Σ.

$$\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v_2}] = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}. \ \boldsymbol{\Sigma} = \begin{bmatrix} 3/\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Step 3. Construct **U**.

$$\mathbf{u}_1 = \frac{1}{\sigma_1}\mathbf{A}\mathbf{v_1} = \frac{1}{3\sqrt{2}}\begin{bmatrix} 2/\sqrt{2} \\ -4/\sqrt{2} \\ 4/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix}, \text{ but } \mathbf{A}\mathbf{v_2} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix};$$

We need two additional orthogonal unit vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ that are orthogonal to $\mathbf{u}_1$. That is, we want to find two vectors $\mathbf{x}$ such that $\mathbf{u}_1^T\mathbf{x} = 0$.

The problem is equivalent to the equation $\mathbf{x}_1 - 2\mathbf{x}_2 + 2\mathbf{x}_3 = 0$. A basis for the solution set is

$$\mathbf{w}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \text{ and } \mathbf{w}_3 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}.$$

Apply Gram-Schmidt process to $\{\mathbf{w}_2, \mathbf{w}_3\}$ to get

$$\mathbf{u}_2 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix} \text{ and } \mathbf{u}_3 = \begin{bmatrix} -2/\sqrt{5} \\ 4/\sqrt{5} \\ 5/\sqrt{5} \end{bmatrix}.$$

Finally, set $\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3]$.

Then $\mathbf{A} = \begin{bmatrix} 1/3 & 2/\sqrt{5} & -2/\sqrt{5} \\ -2/3 & 1/\sqrt{5} & 4/\sqrt{5} \\ 2/3 & 0 & 5/\sqrt{5} \end{bmatrix} \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$

In-class Exercise #16

Q. Find the SVD of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 2 & 1 \end{bmatrix}$

Answer.

Pseudo-inverse

- Ex. (*Reduced SVD and pseudoinverse*)

When Σ contains rows or columns of zeros, a more compact decomposition of **A** is possible. Using the notation established above, let r = rank **A**, and partition **U** and **V** into submatrices whose first blocks contain r columns:

$\mathbf{U} = [\mathbf{U}_r \ \ \mathbf{U}_{m-r}]$, where $\mathbf{U}_r = [\mathbf{u}_1 \ \dots \ \mathbf{u}_r]$ and

$\mathbf{V} = [\mathbf{V}_r \ \ \mathbf{V}_{n-r}]$, where $\mathbf{V}r = [\mathbf{v}_1 \ \dots \ \mathbf{v}_r]$ and

Then $\mathbf{U}_r$ is $m\times r$ and $\mathbf{V}_r$ is $n\times r$. Then partitioned matrix multiplication shows that

$$\mathbf{A} = [\mathbf{U}_r \quad \mathbf{U}_{m-r}]\begin{bmatrix} \mathbf{D} & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \mathbf{V}_r^T \\ \mathbf{V}_{n-r}^T \end{bmatrix} = \mathbf{U}_r\mathbf{D}\mathbf{V}_r^T$$

This factorization of **A** is called a reduced singular value decomposition of **A**. Since the diagonal entries in **D** are nonzero, we can form the following matrix, called the *pseudoinverse* (also, *Moore-Penrose inverse*) of **A**:

$$\mathbf{A}^+ = (\mathbf{U}_r\mathbf{D}\mathbf{V}_r^T)^{-1} = \mathbf{V}_r\mathbf{D}^{-1}\mathbf{U}_r^T.$$

❁ Ex. (*Least-square solution*)
Given the equation **Ax** = **b**, use the pseudo inverse of **A** to define $\hat{\mathbf{x}} = \mathbf{A}^{+}\mathbf{b} = \mathbf{V}_r\mathbf{D}^{-1}\mathbf{U}_r^T\mathbf{b}$. Then, from the *SVD* in Eq. (9),

$$\begin{aligned}\mathbf{A}\hat{\mathbf{x}} &= (\mathbf{U}_r\mathbf{D}\mathbf{V}_r^T)(\mathbf{V}_r\mathbf{D}^{-1}\mathbf{U}_r^T\mathbf{b}) \\ &= \mathbf{U}_r\mathbf{D}\mathbf{D}^{-1}\mathbf{U}_r^T\mathbf{b} \ \ (\because \mathbf{V}_r^T\mathbf{V}_r = \mathbf{I}_r) \ = \mathbf{U}_r\mathbf{U}_r^T\mathbf{b}.\end{aligned}$$

where $\mathbf{U}_r\mathbf{U}_r^T\mathbf{b}$ is the orthogonal projection $\hat{\mathbf{b}}$ of $\mathbf{b}$ onto Col**A**.
Thus $\hat{\mathbf{x}}$ is a least-square solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.

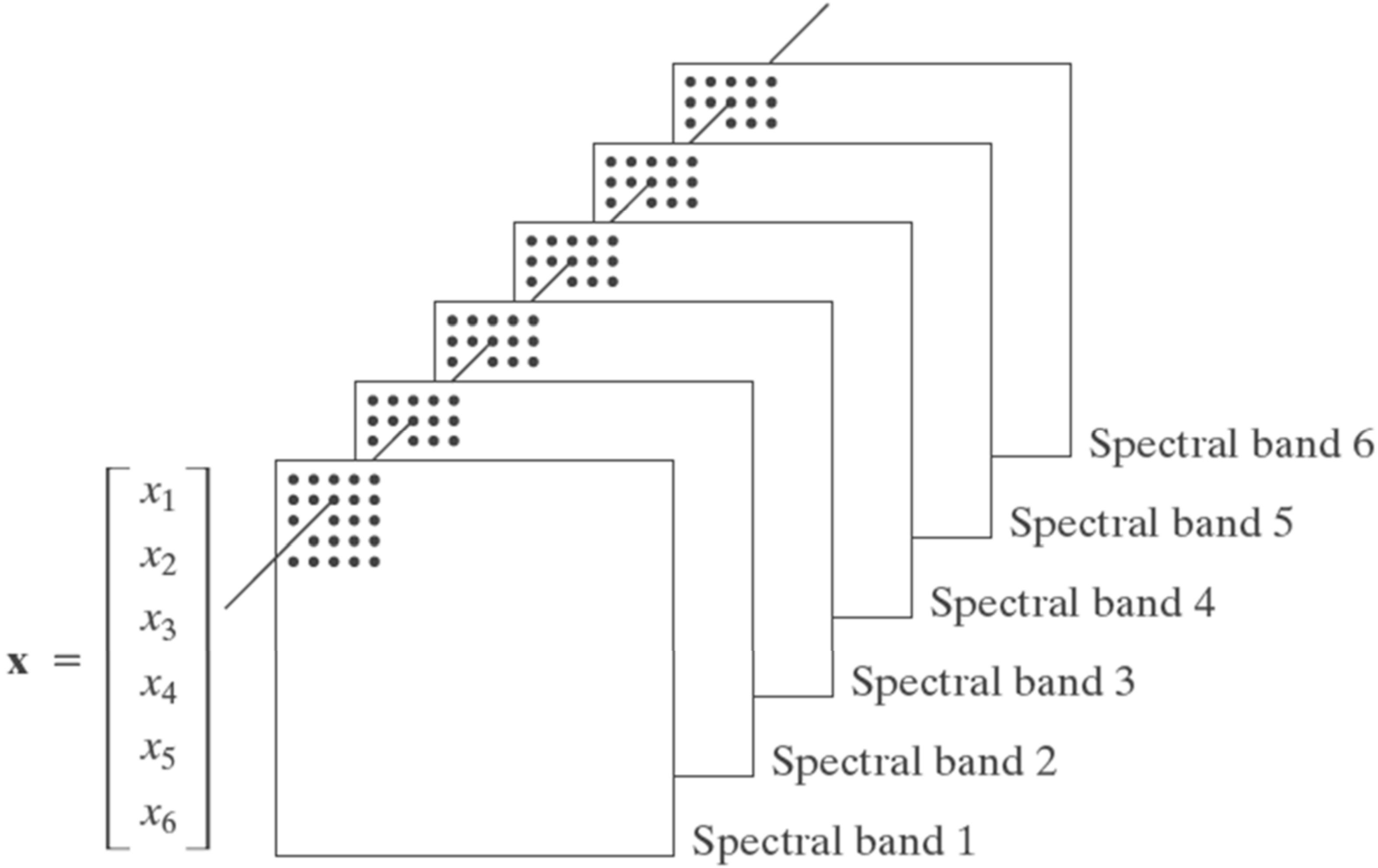
Applications of the singular value decomposition

❁ **The condition number**
Consider the equation **Ax** = **b**. If the singular values of **A** are extremely large or small, round-off errors are almost inevitable, but an error analysis is aided by knowing the entries in Σ and **V**. If **A** is an invertible $n \times n$ matrix, then the ratio σ_1/σ_n of the largest and smallest singular values given the *condition number* of **A**.

Implementation of PCA

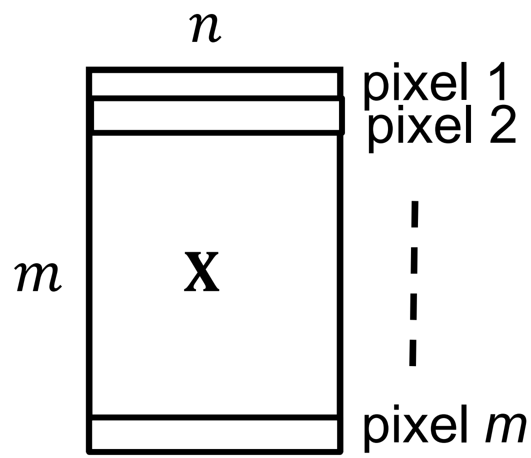
PCA

- Given a multispectral image, treat a pixel having 6 channels as a vector $\mathbf{x} = [x_1 \quad x_2 \quad \cdots \quad x_6]^T$ (random variable).



PCA

Implementation of PCA



- Assume $\bar{\mathbf{x}} = 0$, $\mathbf{C}_x = \frac{1}{n}\mathbf{X}^T\mathbf{X}$,

  where $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m] \in R^{m \times n}$

  m: number of pixels

  n: number of bands

- We can compute the eigenvectors of $\mathbf{C}_x$.
- Or, we can directly perform the **SVD** of $\mathbf{X}$,

  $$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T,$$

  where

  $\mathbf{V}$ is formed by the eigenvectors of $\mathbf{X}^T\mathbf{X}$ and

  $\mathbf{\Sigma}$ is formed by the singular value of $\mathbf{X}$ (σ), that is, $\lambda = \sigma^2$.

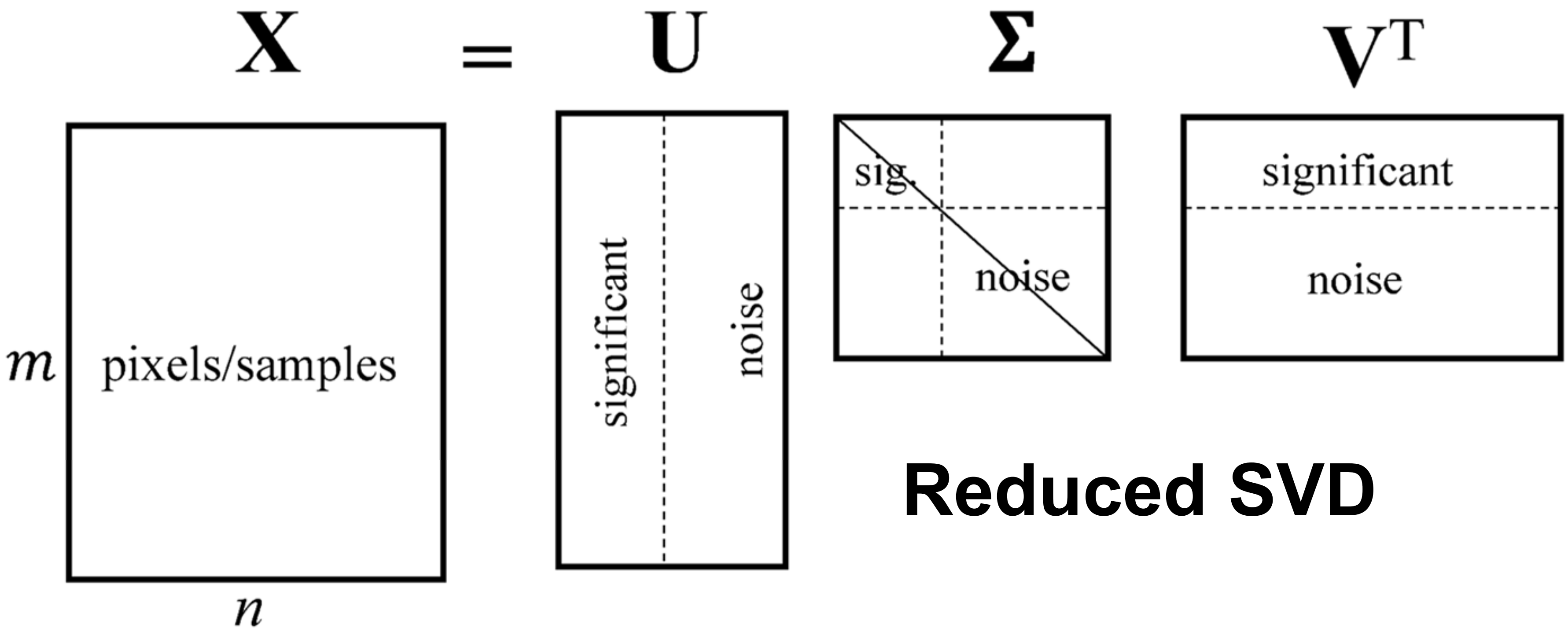
Application of PCA
(Noise Removal and Dimensionality Reduction)

Singular Value Decomposition of the **centered** data matrix **X**.

$$\mathrm{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_m] \in \mathbb{R}^{m \times n},$$

m: number of pixels
n: number of bands

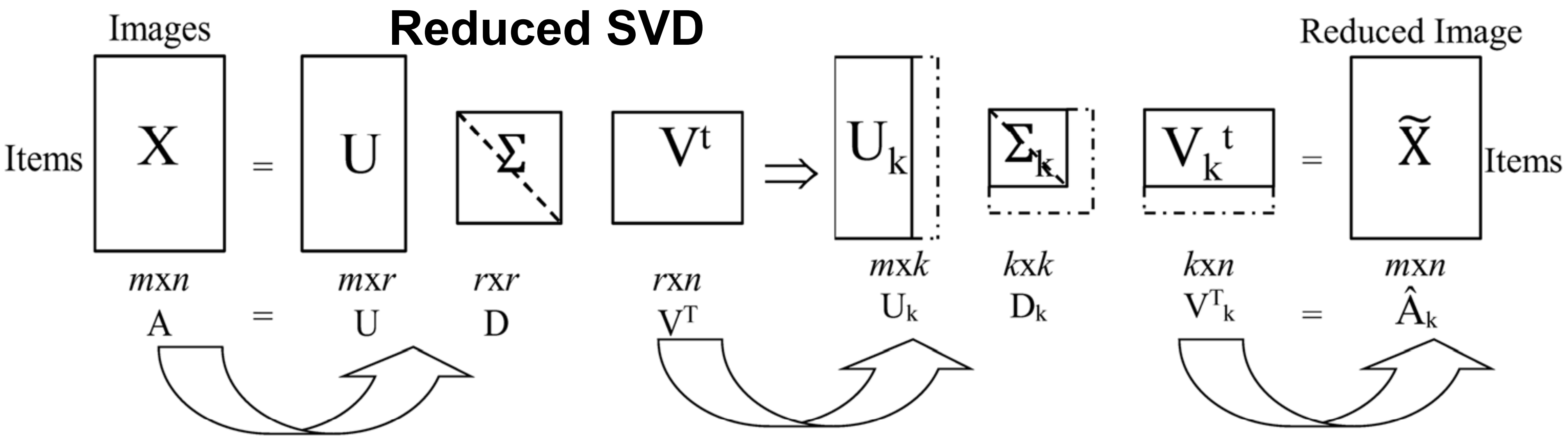
$$\mathbf{X}_{\#\text{pixels} \times \#\text{bands}} = \mathbf{U}\,\boldsymbol{\Sigma}\,\mathbf{V}^{\mathrm{T}}$$



Reduced SVD

Application of PCA

(Noise Removal and Dimensionality Reduction)



Singular Value Decomposition (SVD): Convert term matrix into 3matrices U, Σ and V

Dimensionality Reduction: Throw out low-order rows and columns

Matrix Reconstruction: Multiply to produce approximate term matrix.